

Akshat Khosla

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PROFESSIONAL SUMMARY

Backend and Applied AI Engineer with 2+ years shipping production systems at enterprise scale (.NET, Python, Azure). Delivered 3 end-to-end LLM platforms using RAG, LLM-as-judge, MCP, and distributed tracing on FastAPI and ChromaDB. Published ML researcher (Complex & Intelligent Systems, 2025). Targeting Software Engineer I/II and Applied AI/ML Engineer roles.

TECHNICAL SKILLS

Languages: Python · C# · TypeScript · JavaScript · C/C++ · SQL

Frameworks: FastAPI · Flask · .NET / ASP.NET Core · React · Angular · SQLAlchemy · Pydantic · asyncio · REST APIs

Databases: PostgreSQL · SQL Server · MongoDB · SQLite · ChromaDB · Redis · Query Optimization · Alembic

Cloud & DevOps: Microsoft Azure · Docker · GitHub Actions · CI/CD · Git · Microservices · Distributed Systems · Selenium

AI / ML: RAG · LLM Agents · LLM-as-Judge · Prompt Engineering · MCP · Vector Databases · SentenceTransformers · Anthropic API · spaCy

Practices: System Design · Agile · Event-Driven Architecture · Domain-Driven Design · Distributed Tracing

EXPERIENCE

GEP Software Engineer · Jul 2024 – Present · Hyderabad, India

- Shipped an AI-powered PR Copilot (Claude Opus 4.6) into production, auto-generating test cases and readiness reports from PR descriptions, reducing per-PR review time by ~30% across 8+ engineers and measurably improving defect catch rate.
- Architected a scalable configuration framework across 3+ product modules, cutting feature delivery time by 40% and enabling non-technical self-service workflow customization for enterprise clients.
- Led Angular 8 to 20 migration across 2 production repos with 150+ UI component plugins impacting 10+ website pages, completing all 3 sprints on schedule; co-authored the migration POC with the Platform team and cross-supported additional engineering teams to accelerate org-wide modernization, future-proofing the codebase for AI integration.
- Led end-to-end delivery of a configurable Excel-based data ingestion system, replacing 3 rigid templates and cutting client onboarding time by 60% for new enterprise data types.

Thapar University Research Assistant · Jan 2024 – Jul 2024 · Patiala, India

- Designed a Transformer-augmented Deep Knowledge Tracing model on a 500K+ student interaction dataset, achieving +6.8% AUC and +5.3% F1 over LSTM baselines; research published in Complex & Intelligent Systems (2025).

GEP Technology Intern · May 2023 – Jul 2023 · Mumbai, India

- Engineered a REST API automation script for vulnerability-alert setup across 50+ GitHub repositories, compressing a multi-day manual security task to under 30 minutes (~96% time reduction) and strengthening org-wide compliance posture.

Cadence Design Systems Software Intern · May 2022 – Jul 2022 · Noida, India

- Automated IC waveform data curation and EDA analysis in Python across 100+ simulation datasets, replacing a fully manual evaluation workflow and reducing simulation review time by 50%.

PROJECTS

LLM Evaluation Platform 2026 [GitHub](#)

- Built a multi-judge LLM evaluation platform (LLM-as-judge, faithfulness, relevance, coherence) with asyncio.Semaphore concurrency supporting 100+ concurrent evaluations; provider-agnostic abstraction across Google Gemini and Ollama.
- Shipped FastAPI + PostgreSQL JSONB backend with distributed tracing and query-optimized span retrieval; React + TypeScript dashboard with CI/CD via GitHub Actions + Docker for production-grade availability.

Stack: Python · FastAPI · React · TypeScript · PostgreSQL · SQLAlchemy · Docker · GitHub Actions

MemoryOS: Persistent Memory Layer for LLMs 2026 [GitHub](#) | [Demo](#)

- Implemented a three-tier memory architecture (in-process dict, ChromaDB vector store, SQLite) enabling persistent cross-session recall with sub-100ms retrieval and zero cloud dependencies.
- Shipped an MCP server (JSON-RPC / stdio) compatible with Claude Desktop and Cursor, exposing 6 memory tools backed by a spaCy NER + regex pipeline for automated entity extraction.

Stack: Python · MCP SDK · ChromaDB · SQLite · spaCy · SentenceTransformers · Ollama

DocMind: Document Q&A Agent 2025 [GitHub](#) | [Demo](#)

- Production RAG + LLM agent web app ingesting PDF/DOCX/TXT up to 200 pages, embedding with SentenceTransformers and returning source-attributed answers via a multi-step agentic tool-use loop (Flask, Claude API, ChromaDB).
- Orchestrated a ReAct agent loop with 3 Claude tools, SSE streaming, and a persistent 384-dim HNSW vector index for sub-second cosine-similarity semantic search.

Stack: Python · Flask · Claude API · ChromaDB · SentenceTransformers · RAG · SSE

PUBLICATIONS

Khosla et al. (2025). Knowledge Tracing-Based Intelligent Framework for Formative Assessment of Students' Skills. *Complex & Intelligent Systems*. [\[Link\]](#)

EDUCATION

Thapar University B.E. in Computer Engineering · Aug 2020 – May 2024 · Patiala, India

CGPA: 8.7 / 10